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DS PROGRAMMING

H Harshavardhan
192511171RD

QUESTIONS 10 / 10 completed

AVL Tree- Insert Ele

AVL TREE- INSERT ELEMENTS

"Write a program to create an AVL Tree and insert the following elements:
50, 30, 70, 20, 40, 60, 80.
Construct the AVL Tree, Display the tree using Inorder Traversal, Verify that the height balance property is maintained after each insertion."

PROGRAM EDITOR

C Run Save

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int key, height;
    struct Node *left, *right;
};

int h(struct Node *n) { return n ? n->height : 0; }
int max(int a, int b) { return (a > b) ? a : b; }

struct Node *rotR(struct Node *y) {
    struct Node *x = y->left, *T2 = x->right;
    x->right = y; y->left = T2;
    y->height = max(h(y->left), h(y->right)) + 1;
    x->height = max(h(x->left), h(x->right)) + 1;
    return x;
}

struct Node *rotL(struct Node *x) {
    struct Node *y = x->right, *T2 = y->left;
```

INPUT

Your INPUT goes here!

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C

Run Save

```c
    y->left = x; x->right = T2;
    x->height = max(h(x->left), h(x->right)) + 1;
    y->height = max(h(y->left), h(y->right)) + 1;
    return y;
}

struct Node *insert(struct Node *n, int k) {
    if (!n) {
        struct Node *node = malloc(sizeof(struct Node));
        node->key = k; node->left = node->right = NULL; node->height = 1;
        return node;
    }
    if (k < n->key) n->left = insert(n->left, k);
    else if (k > n->key) n->right = insert(n->right, k);
    else return n;

    n->height = 1 + max(h(n->left), h(n->right));
    int b = h(n->left) - h(n->right);

    if (b > 1 && k < n->left->key) return rotR(n);
    if (b < -1 && k > n->right->key) return rotL(n);
```

INPUT

Your INPUT goes here!

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```c
    if (b > 1 && k < n->left->key) return rotR(n);
    if (b < -1 && k > n->right->key) return rotL(n);
    if (b > 1 && k > n->left->key) { n->left = rotL(n->left); return rotR(n); }
    if (b < -1 && k < n->right->key) { n->right = rotR(n->right); return rotL(n); }
    return n;
}

void inorder(struct Node *r) {
    if (r) { inorder(r->left); printf("%d ", r->key); inorder(r->right); }
}

int main() {
    struct Node *root = NULL;
    int arr[] = {50, 30, 70, 20, 40, 60, 80};
    for (int i = 0; i < 7; i++) {
        root = insert(root, arr[i]);
        printf("Inserted %d | Inorder: ", arr[i]);
        inorder(root); printf("\n");
    }
    return 0;
}
```

INPUT

Your INPUT goes here!